

Tilings

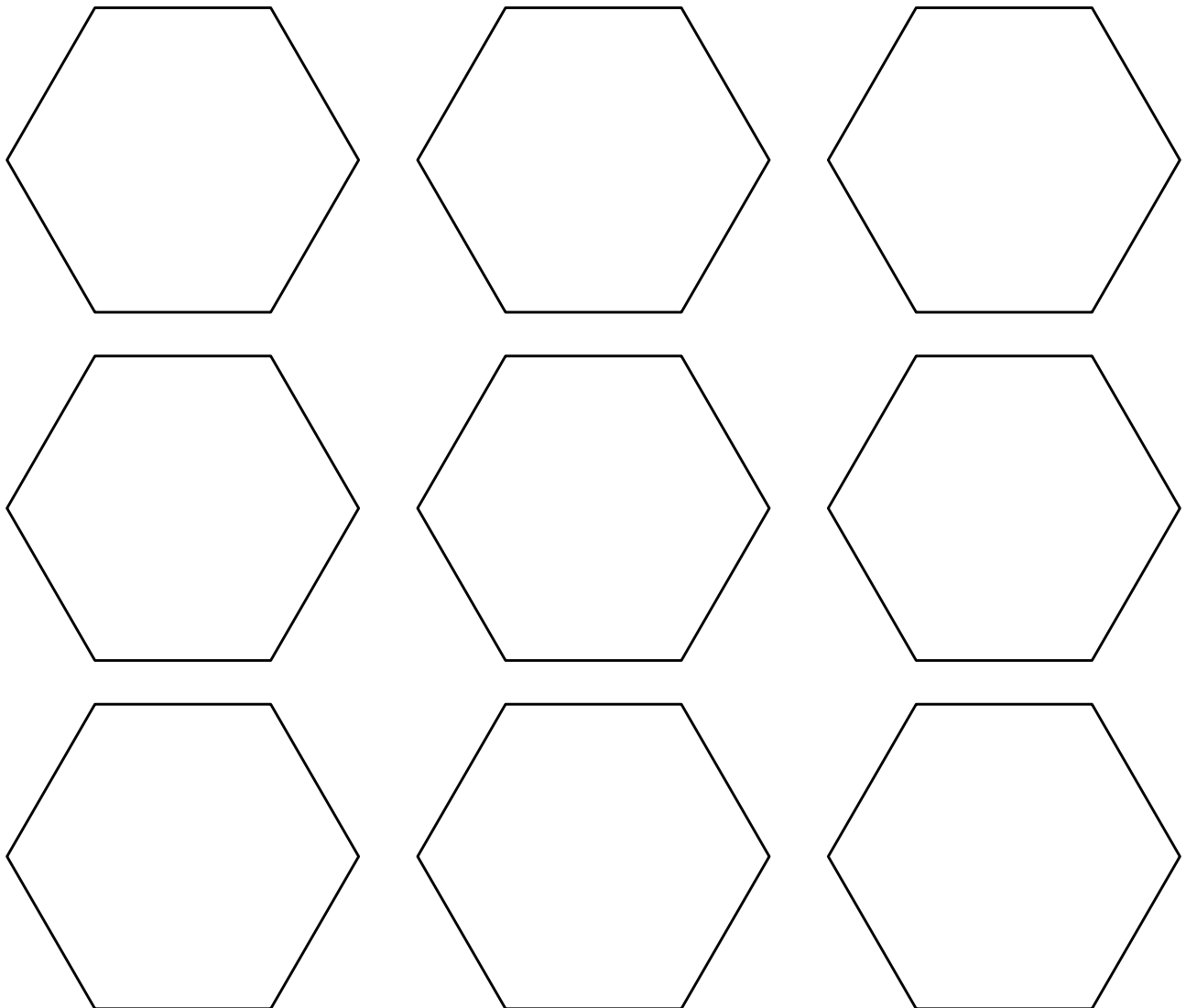
Week 1 · K–1 table

For the grown-up at this table: the shaded boxes are for reading aloud, one at a time. Nothing on these pages needs to be read by the children. Answers, the reasons behind them, and what to do when a puzzle stalls are in the facilitator guide.

1. Fill the hexagon

READ ALOUD

Cover a hexagon with blocks so that it is filled exactly: no gaps, no overlaps, and nothing sticking out over the edge. Then fill the next hexagon a *different* way. How many different ways can you find?

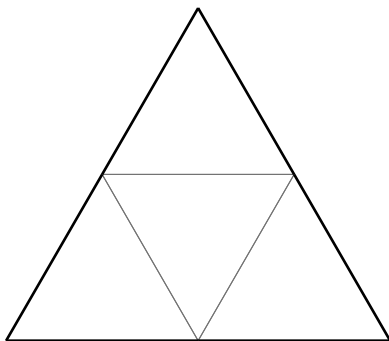


Different ways we found: _____

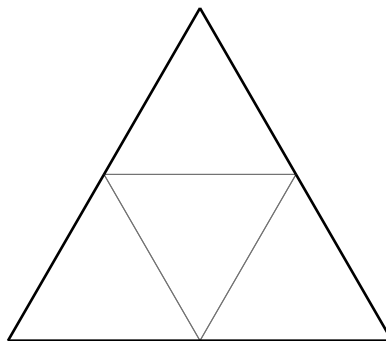
2. The small triangle

READ ALOUD

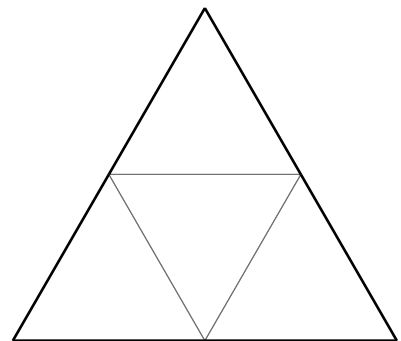
This triangle is made of four little triangles. Count them. First, fill it with green triangles only. How many did you need? Then fill the next one with one red trapezoid and one more block. Which block fits? Then try the last one with blue rhombi only. Keep trying!




green only



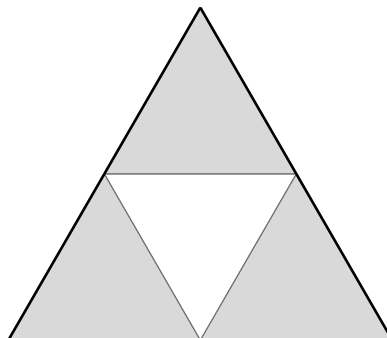
 + ?
one red + one more




blue only

READ ALOUD

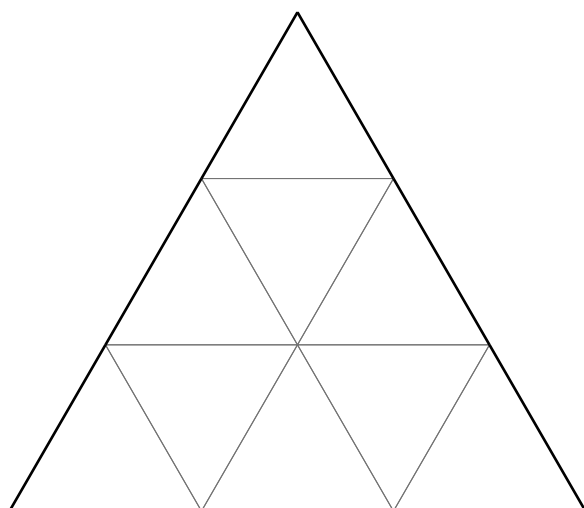
In this picture three little triangles point *up*: they are shaded. One little triangle points *down*: it is white. Put one blue rhombus on the picture, anywhere it fits. What does it cover? Move it somewhere else. What does it cover now? How many shaded triangles are there? How many white ones? Could two blue rhombi ever cover three shaded triangles?



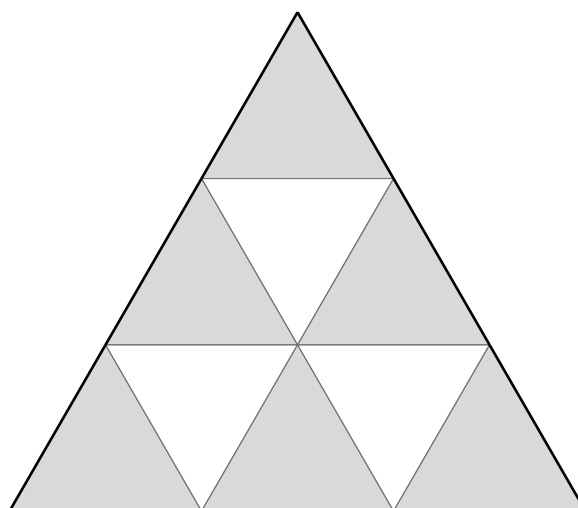
3. The big triangle

READ ALOUD

This bigger triangle is made of nine little triangles. Fill the first one with red trapezoids only. Exactly three will do it, but they have to be placed just right. Then try to fill the second one with blue rhombi only. If it will not work, count: how many shaded (pointing up)? How many white (pointing down)?



red only

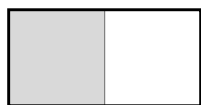


blue only?

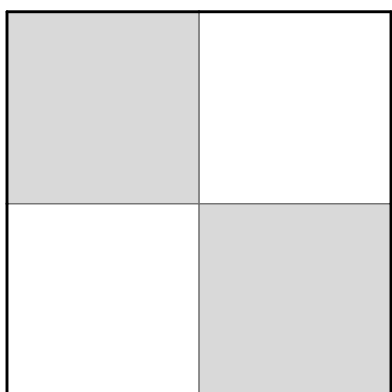
4. Dominoes

READ ALOUD

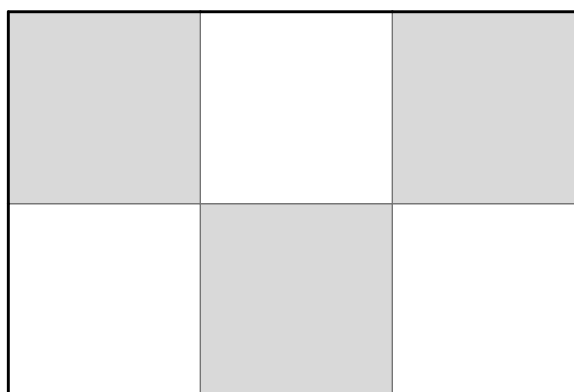
A domino covers two squares: one gray and one white. Cover this square board with dominoes. Now cover the longer board. Can you cover the longer board a different way? How many different ways are there?



a domino



four squares

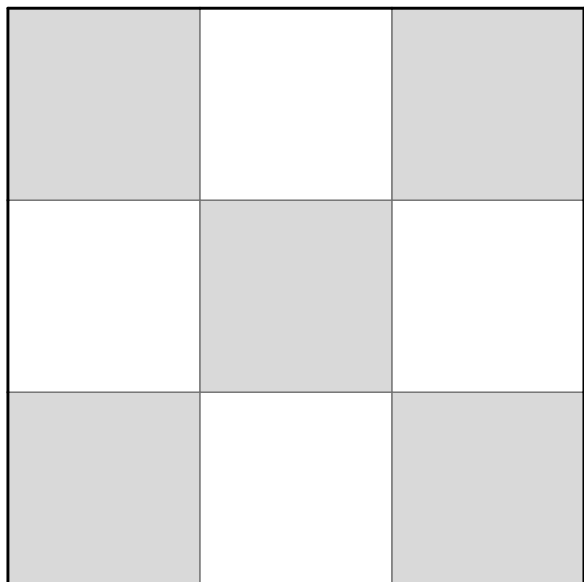


six squares

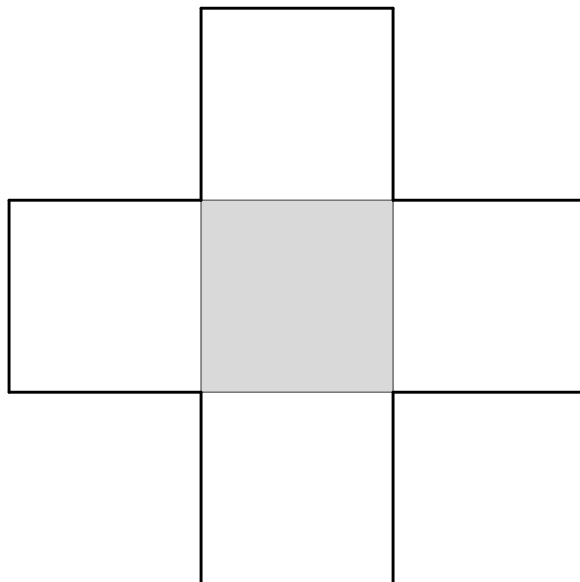
5. Which ones can be covered?

READ ALOUD

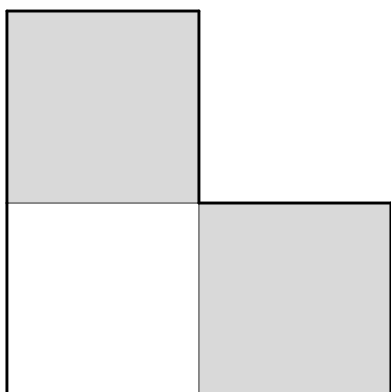
Try to cover each shape with dominoes. Some of them can be covered. Some of them cannot, no matter how you try. When a shape will not work, count its squares. Then count the gray squares and the white squares.



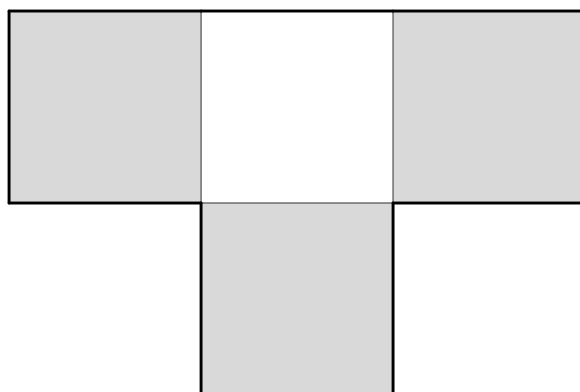
☐ yes ☐ no



☐ yes ☐ no



☐ yes ☐ no

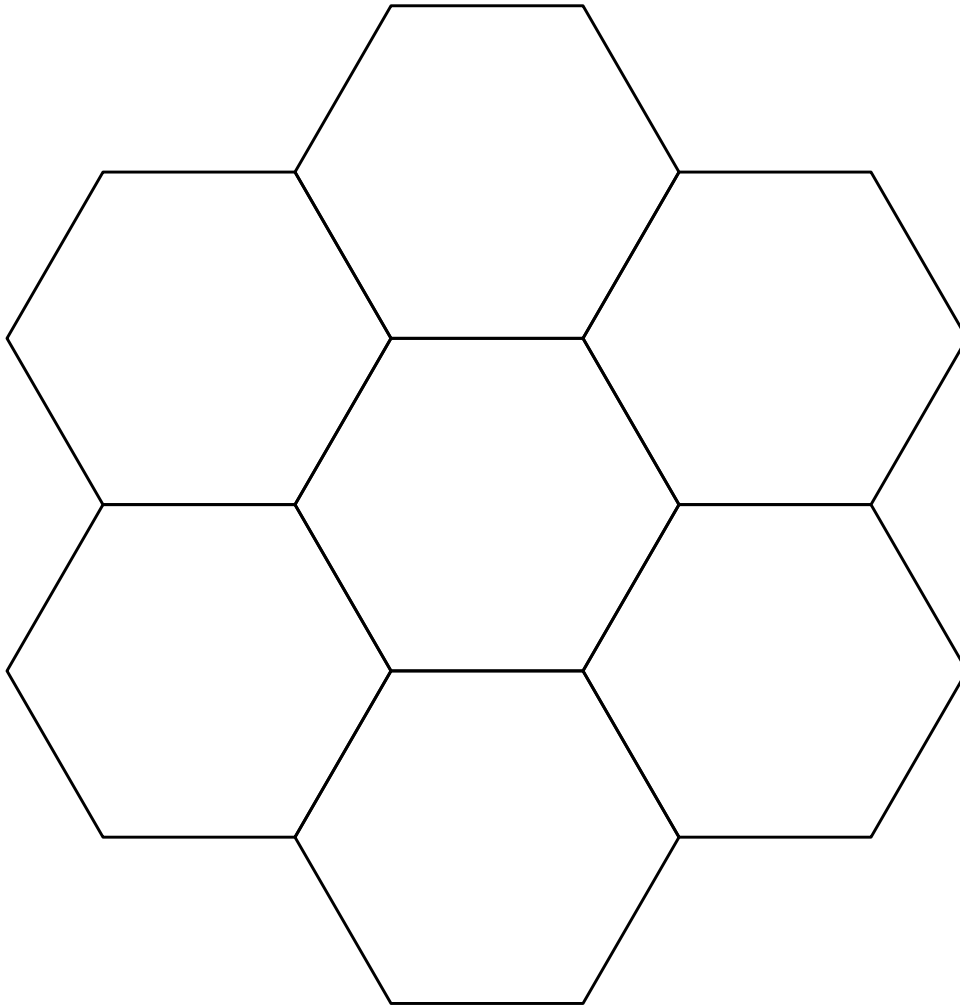


☐ yes ☐ no

6. The flower

READ ALOUD

Fill the whole flower with blocks, any blocks you like. Then count how many blocks you used. Could you fill it with yellow hexagons only? How many would you need? What about red only? Blue only? Green only?



yellow only: _____ red only: _____ blue only: _____ green only: _____

Paper dominoes

Cut along the heavy lines. Each domino is one gray square and one white square.

